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REUNET: AN EFFICIENT REMOTE SENSING IMAGE SEGMENTATION NETWORK

YUHUA WANG[1], YICHEN LIU[2], DEZHAO DENG[3], YINGHUI WANG[4]

[1]Chongqing University of Posts and Telecommunications, Chongqing 400065, China
[2]Southwestern University of Finance and Economics, Chengdu 610074, China
E-MAIL: curley0619@gmail.com

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Abstract:

The extraction of buildings from remote sensing images has long been a prominent research topic in the field of remote sensing image analysis. In this study, we propose a novel approach for building extraction called ReUnet, which combines the residual network and U-net image segmentation network. This model utilizes the skeleton of U-net and the residual units of Resnet to address the issue of gradient diffusion during feature extraction in traditional U-net networks. The residual structure of ReUnet simplifies the training of deep networks and promotes information propagation. We evaluate our network using the LEVIR-CD dataset and compare it with three advanced deep learning-based methods for remote sensing building extraction, namely PSnet, U-net, and Resnet. The proposed method outperforms all compared methods in terms of its ability to accurately extract buildings. Overall, our results suggest that ReUnet is a promising approach for building extraction from remote sensing images.

Keywords:

Image segmentation; Surface structure identification; Residual network; Unet; Skip connection

1. Introduction

In recent years, remote sensing technology has been widely utilized in various domains, significantly driving the development of aerial image processing. Landmark building identification from aerial imagery is of paramount importance for detecting changes in buildings across the country through remote sensing data. To achieve this, image segmentation is a fundamental task in remote sensing that needs to be accurately accomplished. However, the extraction of building features from remote sensing imagery remains a challenging endeavor due to the presence of noise, occlusion, and lighting variations in the original background. Consequently, the development of advanced image segmentation methods is essential to address these challenges[1].

In recent years, various methods have been proposed for the extraction of surface buildings from remote sensing images, which can be broadly categorized into two major types: traditional statistical learning theories and modern deep learning methods. Traditional methods mainly include morphological[2] or edge detection[3] algorithms that have evolved into machine learning methods such as random forests[4] and support vector machines[5]. However, traditional methods often perform poorly in recognizing shadows, while deep learning-based methods are well-suited to adapting to such anomalies. The most advanced methods for semantic segmentation include fully convolutional neural networks[6], as well as residual neural networks (ResNet)[7] and Unet[8].Inspired by deep residual learning and U-Net, we propose the Deep Residual U-Net in this paper, which is an architecture that leverages the advantages of deep residual learning and the U-Net architecture. Our proposed Deep Residual U-Net (ReUNet) is built upon the U-Net architecture. There are two aspects of differences between our deep ReUNet and U-Net. Firstly, we use residual units as basic blocks instead of regular neural units to construct the deep ResUNet. Secondly, cropping operations are unnecessary and therefore removed from our network, resulting in a more elegant architecture and better performance.

This paper presents a methodology that employs an integrated residual neural network and Unet image segmentation network for the recognition of surface buildings in remote sensing images. Building change detection is a key research focus in the field of remote sensing, and this paper emphasizes the extraction of buildings from remote sensing images. The remaining sections of the paper are organized as follows: Section

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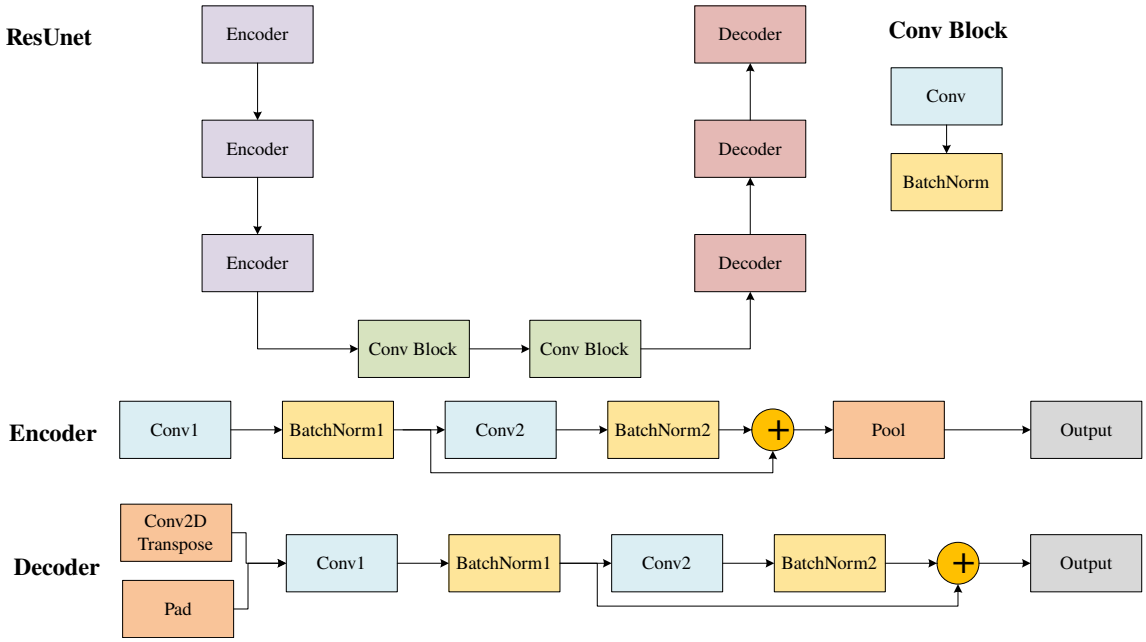


FIGURE 1. The detailed architecture of the proposed ReUnet.

2 presents a detailed description of the Unet and Resnet architectures, as well as the proposed ReUnet framework. Section 3 outlines the overall processing workflow for remote sensing surface building recognition using the LEVIR-CD dataset. Section 4 summarizes the advantages of the proposed framework and highlights future directions for research and development.

2. The ReUnet model

2.1. Unet

UNet is a variant of fully convolutional neural networks, whose network structure is symmetric and resembles the letter U. Each layer structure is composed of some two-dimensional planes, and each plane is composed of multiple different and independent neurons[9]. This is different from fully connected neural networks, but their overall architecture is similar. It consists of two parts: an encoder and a decoder. The encoder is responsible for gradually downsampling the input image and extracting feature representations of different scales, while the decoder gradually upsamples the feature maps of the encoder and fuses them with the corresponding feature maps of the encoder to obtain a prediction segmentation result of the same size as the original input image[10].

Semantic segmentation requires the use of low-level details in addition to preserving high-level semantic information to achieve better segmentation results. However, training such a deep neural network is very challenging, especially with limited training samples. One way to address this problem is to use a pre-trained network and fine-tune it on the target dataset, as done in the present work. Another method is to adopt extensive data augmentation, as done in UNet. Besides data augmentation, we believe that the architecture of UNet also helps alleviate training problems. The intuition behind this is that replicating low-level features into corresponding high-level features actually creates a path for information propagation, allowing signals to propagate more easily between low-level and high-level features. This not only benefits backpropagation during training, but also compensates for low-level details in high-level semantic features. This is somewhat similar to the idea behind residual neural networks. In this paper, we demonstrate that the performance of UNet can be further improved by replacing ordinary units with residual units.

2.2. Resnet

The deeper the multi-layer neural network, the better its performance, but it may hinder training and lead to degradation problems. To overcome these issues, residual neural networks

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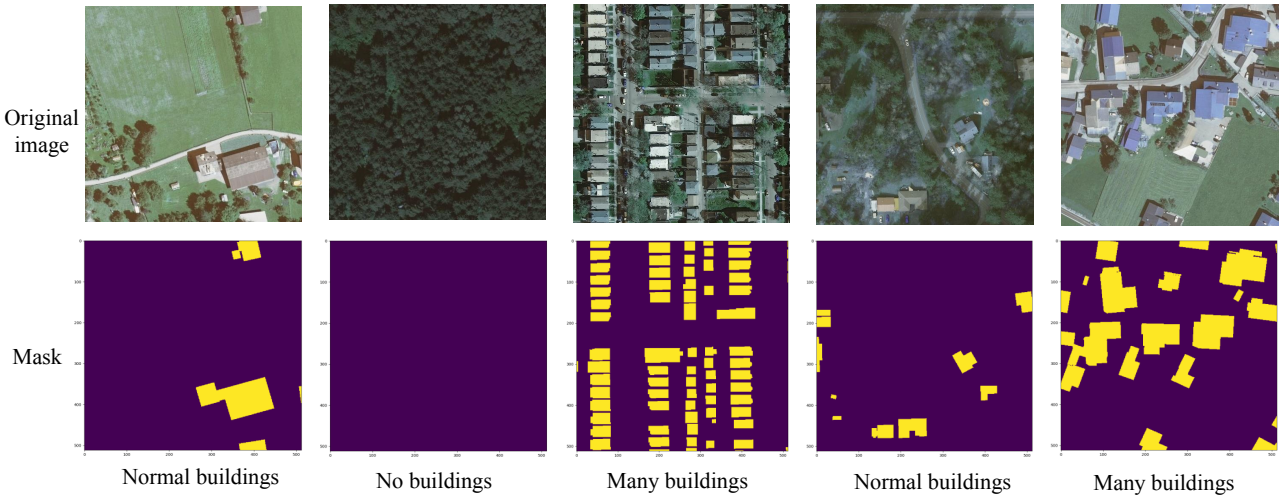


FIGURE 2. Part of the dataset.

were proposed to facilitate training and solve the degradation problem. Residual neural networks are composed of a series of stacked residual units. Each residual unit can be represented in a general form:

$$\begin{aligned} y_l &= h(x_l) + F(x_l, W_l) \\ x_{l+1} &= f(y_l) \end{aligned} \tag{1}$$

In this passage, x_l and x_{l+1} are the input and output of the l-th residual unit, $F(\cdot)$ is the residual function, $F(y_l)$ is the activation function, and h(xl) is the identity mapping function, typically $h(x_l) = x_l$. There are multiple combinations of batch normalization (BN), ReLU activation, and convolution layers in residual units.

2.3. ReUnet

In this paper, we propose the deep ReUnet, a semantic segmentation neural network that combines the strengths of both U-Net and residual neural networks. This combination yields two advantages:

1) the residual units simplify network training;

2) the skip connections within residual units and between low-level and high-level parts of the network facilitate information propagation without degradation.

This allows for the design of a neural network with fewer parameters, while still achieving good performance on semantic segmentation.

The convolutional neural network structure established in this paper is shown in Fig.1, which adopts a symmetric encoder-decoder structure consisting of three encoders, two convolution blocks, and three decoders. The network includes three sub-modules: encoder, decoder, and convolution block. The encoder defines four convolutional layers (referred to as down1, down2, down3, and down4), the decoder defines four convolutional layers (referred to as up4, up3, up2, and up1), and the convolution block defines two convolutional layers (referred to as mid conv1 and mid conv2) for further feature extraction. The feature extraction of the image is obtained by processing through the convolutional and pooling layers.

Encoder module defines the structure of the encoder, including two convolutional layers, two normalization layers, and one max-pooling layer, where one convolutional layer and one normalization layer are combined as a residual block with skip connection. During initialization, input and output channel numbers are given, and then the convolutional and normalization layers are created using Conv2D and BatchNorm for downsampling the image size. During the forward pass, the input data goes through conv1 and bn1 layers, then conv2 and bn2 layers, and the resulting feature map from the residual block is added to the input feature map with skip connection. Then the ReLU activation function is applied for nonlinear transformation, and finally, the feature map is downsampled using the pooling layer, returning both the downsized and original feature maps[11].

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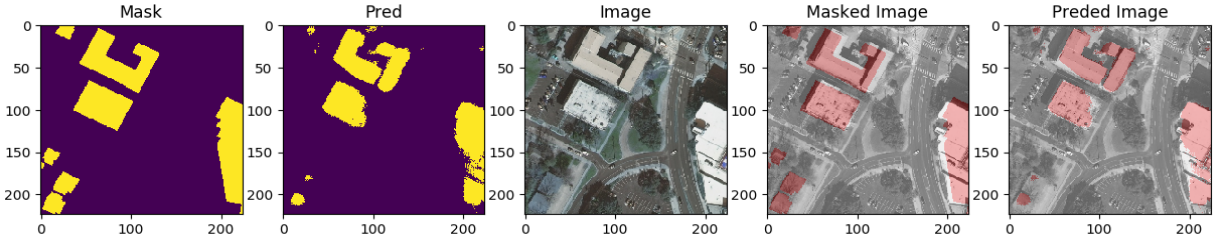


FIGURE 3. Results of Unet image segmentation.

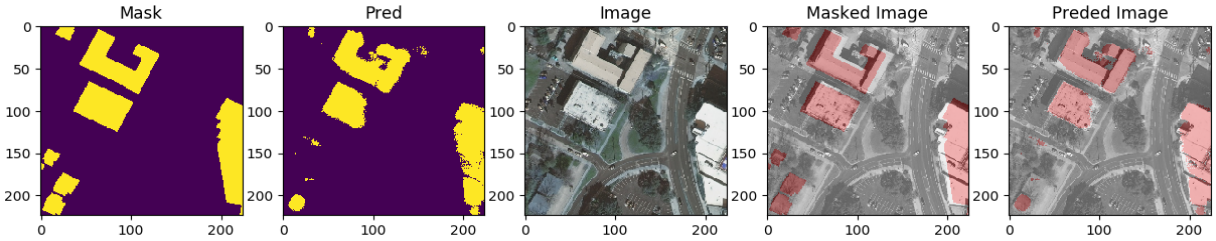


FIGURE 4. Results of ReUnet image segmentation.

Decoder module defines the structure of the decoder, including one transposed convolutional layer, two convolutional layers, and two normalization layers, where one convolutional layer and one normalization layer are combined as a residual block with skip connection. During initialization, input and output channel numbers are given, and then the upsampling layer is created using Conv2D Transpose, and convolutional and normalization layers are created using Conv2D and BatchNorm for upsampling the feature map to the original size. During the forward pass, the input data goes through the transposed convolutional layer, then is padded using Pad2D to match the size of the input data. Then the upsampled feature map and the original feature map are concatenated, and the resulting feature map from the residual block is added to the input feature map with skip connection. Then the ReLU activation function is applied for nonlinear transformation, and the final output is the decoded feature map.

After defining the network architecture, it is necessary to define the accuracy and loss functions. Based on the characteristics of the labels, we can transform the classification of the entire image into the classification of individual pixels(binary classification, whether it is a building or not). In this case, 1 is used to represent a building, and 0 is used to represent other backgrounds. As this is a binary classification problem, binary cross-entropy loss function is used in this paper. The formula for the binary cross-entropy loss function is as follows[12]:

$$L = -[y * log(\hat{y}) + (1 - y) * log(1 - \hat{y})] \tag{2}$$

Here, L represents the loss function, y represents the true label, and $\hat{y}$ represents the predicted label. X is the predicted result, and Y is the true label result. When X is identical to Y, the Dice coefficient is equal to 1. The ranking list uses the average Dice coefficient of all test set images to measure the accuracy, with a higher score indicating better performance.

3. Experiment

3.1. Datasets

TABLE 1. LEVIR-CD dataset.

	training set	test set
Labels	24796	/
No Labels	5204	2500

LEVIR-CD[13] is a large-scale remote sensing dataset for building change detection, consisting of 637 pairs of ultra-high-resolution image patches of size 1024×1024 pixels, including 30,000 training set images and 2,500 test set images annotated with binary labels (1 for change and 0 for no change). With various types of buildings, such as villas, high-rise apartments,

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small garages, and large warehouses, Levir-CD covers a significant time span of 5 to 14 years, showing land use changes and especially building growth, making it ideal for training algorithms to learn deeply. Therefore, this dataset is chosen for experimentation in this paper.

The specific data set distribution is shown in TABLE 1. Fig.2. shows a representative dataset with a lot of buildings; No building; And the original map of the normal building distribution and the labels that identify the buildings.

3.2. Evaluation and Prediction

The above improvement to the model has increased its accuracy, and next we will evaluate and predict using the model.

To evaluate and predict the improved UNet model on the LEVIR-CD dataset, we first need to define similarity and then use instructions to make predictions. When using the model for prediction, the input data is no longer the training sample but the test sample. By running the results, the test accuracy is 76.52%, which is similar to the accuracy of the improved model on the training set in this paper. Then we use the improved model for prediction and visualize the results, as shown in the figure below:

It can be observed that the ReUnet proposed in this paper has a deeper network structure, stronger model representation ability, accelerated training convergence compared to traditional segmentation networks. Moreover, it also results in more accurate data prediction.

3.3. Comparison with other methods

As shown in Fig.5 and Fig.6, it can be seen that during the training process, the loss value periodically decreases, indicating that the training is effective, and the accuracy periodically increases. From the visualization curve of the loss value, it can be seen that the loss value has not yet reached a convergent state, and there is still some room for improvement. As shown in Table 2, there is still potential for further improvement in accuracy.

The above algorithm obtained a loss function value of 0.5946 and an accuracy of 72.28%. This paper compared the representation effect of different network structures on the segmentation model.

From the experimental results in TABLE 2, it can be seen that the UNet with residual blocks has a higher accuracy than the UNet without residual blocks, which also indicates that introducing residual structures can indeed improve the model's representation ability.

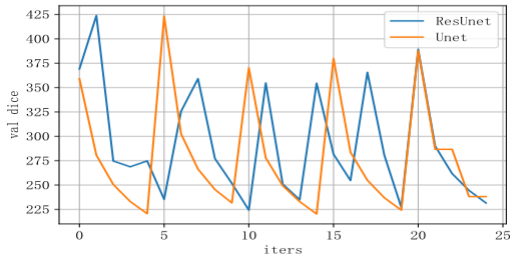


FIGURE 5. Dice visualization of the validation set.

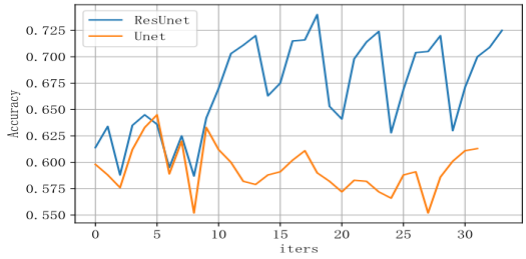


FIGURE 6. Accuracy of the validation set.

At the same time, it can be seen from TABLE 2 that compared with PSPNet, the UNet structure has better segmentation results in this task.

4. Conclusion

In this paper, we present ReUnet, a deep learning network for building extraction in remote sensing images that leverages the benefits of residual learning and U-Net. The residual units and skip connections between encoding and decoding paths facilitate forward and backward information propagation, resulting in improved performance. ReUnet's simplicity and superior performance demonstrate the potential of designing compact and effective neural networks. Compared to U-Net and two other state-of-the-art deep learning-based methods for re-

TABLE 2. Dice coefficient and jaccard index values for different method metrics.

Method	Dice coefficient	Jaccard
PSPnet	0.2111	0.1482
Unet	0.6373	0.4814
ReUnet	**0.7228**	**0.5946**

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mote sensing building image segmentation, ReUnet achieves comparable or better performance with only one-quarter of the parameters.

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